

CrediFlow Documentation

1. CustomerController (api/customer)

Routes & Endpoints

1. GET api/customer/all
 - Returns a list of all customers.
 - Response: List of CustomerDTO.
2. POST api/customer/create
 - Creates a new customer.
 - Request Body: CustomerDTO (**Customer_Id, Status, Credit_Score are auto generated**)

```
{
  "Name": "John Doe",
  "Address": "123 Main St",
  "Email": "john.doe@example.com",
  "Gender": 1.0,
  "Card_Number": "1234-5678-9012-3456",
  "Monthly_Income": 50000.0
}
```
 - Response: Created CustomerDTO.
3. GET api/customer/customer_loans/{id}
 - Gets all loans for a specific customer.
 - URL Parameter: id (Customer_Id)
 - Response: List of CustomerLoanDTO.
4. GET api/customer/customer/{id}
 - Gets details of a specific customer.
 - URL Parameter: id (Customer_Id)
 - Response: CustomerDTO.
5. POST api/customer/update
 - Updates an existing customer.
 - Request Body: CustomerDTO (**can be updated with partial values, but id must be entered**)
 - Response: Updated CustomerDTO.
6. POST api/customer/delete/{id}
 - Deletes a customer.
 - URL Parameter: id (Customer_Id)
 - Response: Success or error message.

Data Format

7. CustomerDTO
 - Customer_Id (int)
 - Name (string)
 - Address (string)
 - Email (string)
 - Credit_Score (float)
 - Gender (float)
 - Card_Number (string)
 - Monthly_Income (float)

- Status (string)

2. LoanController (api/loan)

Routes & Endpoints

8. GET api/loan/all
 - Returns all loans.
 - Response: List of LoanDTO.
9. GET api/loan/{id}
 - Gets details of a specific loan.
 - URL Parameter: id (Loan_Id)
 - Response: LoanDTO.
10. POST api/loan/create
 - Creates a new loan.
 - Request Body: (Loan_Id and Installment_Amount is calculated by system)


```
{
    "Loan_Type": "Personal",
    "Loan_Duration_Months": 12,
    "Interest_Percentage": 5.5,
    "Penalty_Percentage": 2.0,
    "Loan_Amount": 12000,
    "Minimum_Credit_Score": 600.0
  }
```
 - Response: Created LoanDTO.
11. GET api/loan/Get_eligible_loans/{customer_id}
 - Gets loans eligible for a specific customer.
 - URL Parameter: Customer_id
 - Response: List of LoanDTO.
12. POST api/loan/update
 - Updates an existing loan.
 - Request Body: LoanDTO (can be updated with partial values, but id must be entered)
 - Response: Updated LoanDTO.
13. POST api/loan/delete/{id}
 - Deletes a loan.
 - URL Parameter: id (Loan_Id)
 - Response: Success or error message.

Data Format

14. LoanDTO
 - Loan_Id (int)
 - Loan_Type (string)
 - Loan_Duration_Months (int)
 - Interest_Percentage (float)
 - Penalty_Percentage (float)
 - Installment_Amount (float)

- **Loan_Amount (int)**
 - **Minimum Credit Score (float)**
-

3. CustomerLoanController (api/customerloan)

Routes & Endpoints

15. **POST api/customerloan/create** (Customer getting a loan)
 - Creates a new customer loan.
 - Request Body: CustomerLoanDTO (**just enter Customer_Id and Loan_Id**)
 - Response: Created CustomerLoanDTO.
16. **GET api/customerloan/get/{id}**
 - Gets details of a specific customer loan.
 - URL Parameter: id (Customer_Loan_Id)
 - Response: CustomerLoanDTO.
17. **GET api/customerloan/all**
 - Returns all customer loans.
 - Response: List of CustomerLoanDTO.
18. **GET api/customerloan/active_loans**
 - Returns all active customer loans.
 - Response: List of CustomerLoanDTO.
19. **GET api/customerloan/closed_loans**
 - Returns all closed customer loans.
 - Response: List of CustomerLoanDTO.
20. **POST api/customerloan/update**
 - Updates an existing customer loan.
 - Request Body: CustomerLoanDTO (**can be updated with partial values, but id must be entered**)
 - Response: Updated CustomerLoanDTO.
21. **POST api/customerloan/delete/{id}**
 - Deletes a customer loan.
 - URL Parameter: id (Customer_Loan_Id)
 - Response: Success or error message.

Data Format

22. CustomerLoanDTO
 - Customer_Loan_Id (int)
 - Customer_Id (int)
 - Loan_Id (int)
 - Loan_Taken_Date (DateTime)
 - Outstanding_Amount (float)
 - Next_Installment_Date (DateTime)
 - Next_Installment_Amount (float)
 - Total_Paid_Amount (float)
 - Status (string)
 - Loan_End_Date (DateTime)
 - LoanDTO (object)
 - CustomerDTO (object)

4. PaymentController (api/payment)

Routes & Endpoints

- 23. POST **api/payment/LoanReturn** (monthly installment returns)
 - Processes a loan payment (installment).
 - Request Body: **PaymentDTO** (only requires **Customer_Id** and **Customer_Loan_Id**)
 - Response: Updated **CustomerLoanDTO** or error message.
- 24. GET **api/payment/all**
 - Returns all payments.
 - Response: List of **PaymentDTO**.
- 25. GET **api/payment/{id}**
 - Gets details of a specific payment.
 - URL Parameter: **id (Payment_Id)**
 - Response: **PaymentDTO**.
- 26. POST **api/payment/create**
 - Creates a new payment.
 - Request Body: **PaymentDTO**

```
{
  "Payment_Id": 1,
  "Payment_Date": "2025-10-12T10:30:00",
  "Customer_Id": 1,
  "Customer_Loan_Id": 1,
  "Amount": 1000
}
```
 - Response: **Created PaymentDTO (Bool)**.
- 27. POST **api/payment/update**
 - Updates an existing payment.
 - Request Body: **PaymentDTO** (can be updated with partial values, but **id must be entered**)
 - Response: Updated **PaymentDTO**.
- 28. POST **api/payment/delete/{id}**
 - Deletes a payment.
 - URL Parameter: **id (Payment_Id)**
 - Response: Success or error message.

Data Format

- 29. **PaymentDTO**
 - **Payment_Id** (int)
 - **Payment_Date** (DateTime)
 - **Customer_Id** (int)
 - **Customer_Loan_Id** (int)
 - **Amount** (float)
 - **CustomerDTO** (object)
 - **CustomerLoanDTO** (object)

5. NotificationController (api/notification)

Routes & Endpoints

- 30. GET **api/notification/all**
 - Returns all notifications.
 - Response: List of NotificationDTO.
- 31. GET **api/notification/{id}**
 - Gets details of a specific notification.
 - URL Parameter: **id (Notification Id)**
 - Response: NotificationDTO.
- 32. GET **api/notification/customer/{id}**
 - Gets all notifications for a specific customer.
 - URL Parameter: **id (Customer_Id)**
 - Response: List of NotificationDTO.
- 33. POST **api/notification/create**
 - Creates a new notification.
 - Request Body: NotificationDTO (**Date, Is_Read and CustomerDTO is automatically added**)

```
{
  "Notification_Id": 1,
  "Title": "Loan Issued",
  "Message": "Your loan has been successfully issued.",
  "Customer_Id": 1,
  "Is_Read": false,
}
```
 - Response: Success or error message.
- 34. POST **api/notification/delete/{id}**
 - Deletes a notification.
 - URL Parameter: **id (Notification Id)**
 - Response: Success or error message.

Data Format

- 35. NotificationDTO
 - Notification_Id (int)
 - Title (string)
 - Message (string)
 - Date (DateTime)
 - Customer_Id (int)
 - Is_Read (bool)
 - CustomerDTO (object)

Key Features

- A. Customers can take loans.
- B. Customers can return loans on monthly installments.
- C. Introduction of Credit Score that is calculated from customer's monthly income
- D. Every customer is set a status based on his/her's monthly income.
- E. Customer can only access loans based on there credit score.
- F. If a customer's credit score is in range of 200-299 he becomes restricted user, in order to continue loan service he have re-activated the account by paying reactivation fee (Loan_type = Re-Activation Loan) within 30 days else he would be terminated.
- G. If a customer's credit score becomes below 200 or delays to pay re activation fee, his/her's account would be terminated and they can't use that account ever.
- H. The credit score is gradually incremented by 20 for per on time installment payment.
- I. The credit score is also gradually decremented by 25 for per late payment.
- J. Automated late payment detection and late fee addition.
- K. While creating a Loan, the monthly installment is calculated from the compound interest formula.
- L. Automated notification is created based on task.
- M. Check if customer has read a notification or not.
- N. Automated mails are sent based on task.
- O. Automated reminder is set for (7 days, 3 days, dealine day).

Credit score and status table:

Score range	Rating	Meaning
<200	Terminated	Can't use this account ever again
200-299	Restricted	Must pay activation fee
300-499	Poor	Very risky
500-649	Fair	Moderate
650-749	Good	Reliable
>750	Best	Very low risk

Credit score based on salary table

Salary Range	Credit Score
<15000	350
15000-29999	450
30000-49999	500
50000-79999	600
80000-120000	650

Database Schema:

